Tutorial 4

Indian Institute of Technology Jodhpur
Mathematics-I (MAL1010)

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1. Define the sequence $\{x_n\}$ by
$$x_{n+1} = \sqrt{2 + x_n}\,, \qquad x_1 \in [0, 2].$$
   (a) Show that $\{x_n\}$ is bounded and monotone increasing.
   (b) Prove that the sequence converges and find $\lim_{n\to\infty} x_n$.

2. Let $\{x_n\}$ be a bounded sequence with $|x_n| \le M$ for all n. Define the averages
$$a_n = \frac{1}{n}\sum_{k=1}^{n} x_k.$$
   (a) Show that $\{a_n\}$ is bounded.
   (b) If $x_n \to L$, prove that $a_n \to L$.

3. Consider the sequence
$$x_n = (-1)^n + \frac{1}{n}, \qquad n \ge 1.$$
   (a) Show that $\{x_n\}$ is bounded.
   (b) Show that $\{x_n\}$ is not monotone.
   (c) Determine the subsequential limits.

4. Consider the sequence $\{x_n\}$ defined by $x_n = \sin n, \qquad n \in \mathbb{N}.$
   (a) Show that $\{x_n\}$ is bounded.
   (b) Show that $\{x_n\}$ is not monotone.
   (c) Determine the set of subsequential limits, and compute $\limsup x_n$ and $\liminf x_n$.

5. Find the limit of these sequences,
   (a) $\{\frac{3n^2+2n+1}{n^2+1}\}$.
   (b) $(\sqrt{n+1} - \sqrt{n})$.

6. Determine whether $a_n = \ln(n)$ diverges or convergence.

7. Show that $b_n = (1 + \frac{1}{n})^{n^2}$ is divergent.

8. Given $a_n = (-1)^n . \frac{n}{n+1}$.
   (a) Find all subsequences that converge.
   (b) What are their limits?

9. Prove that every bounded sequence of real numbers has a convergent subsequence.